

Business or Mission Analysis

AI-Assisted Stock Research
and Ranking Platform (StockLens)

Group 29

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1 Introduction

A beginner comparing two stocks can quickly end up moving among price charts, company records, news stories, and informal advice. StockLens is a proposed browser-based research platform designed to bring that work into one clear path for beginning retail investors and students. A user selects stocks, the platform retrieves dated market and news evidence, and a documented scoring method produces a comparison. The dashboard then shows the ranking, the factors that drove it, the supporting sources, and a plain-language explanation. The goal is not simply to display more information; it is to help the user follow how the evidence became a result and to make the same comparison again.

The user remains responsible for reviewing the evidence and making investment decisions. Automated trading, brokerage connections, personalized portfolio management, and claims about future returns are outside the initial system boundary.

1.1 Description of Business or Mission Analysis

Business or mission analysis begins with the reason for the system rather than with a preferred technology. It asks what problem users face, who is affected, what the system must accomplish, where its boundary lies, and how it would operate over its life cycle. Answering those questions first connects the StockLens idea to a real user need before detailed requirements or design choices are fixed.

For StockLens, this report identifies the main user problems and opportunities, stakeholder needs, assumptions, and proposed measures of effectiveness. It also defines the operational concept, primary use case, action flow, system inputs and outputs, preliminary life cycle concepts, and alternative solution classes. Together, these elements define the intended foundation for the Innoslate model and later system requirements; the consistency review records the shared model baseline.

1.2 Purpose of Business or Mission Analysis

The purpose of this analysis is to give the team a shared mission baseline for StockLens. In practical terms, two team members should be able to describe the same user, the same system boundary, and the same path from a research question to an inspectable result. The intended outcome is a repeatable stock comparison in which the selected stocks, analysis date, scoring factors, main drivers, and supporting sources stay visible.

The proposed ranking is produced through defined calculations that can be repeated and checked. AI is intended to explain relevant evidence in plain language; it does not independently control the numerical ranking or replace the user's judgment. The report therefore explains the selected system concept and identifies how it should later be evaluated.

1.3 Implementation and Prototype Status

This Business and Mission Analysis establishes the conceptual baseline, operational architecture, and proposed validation metrics for the StockLens platform. It represents the requirements and design phase of the systems engineering lifecycle. At this stage, the models define the intended system behavior, and no functional software prototype has been implemented for live demonstration. A local Walking Skeleton uses fixed data and service stubs only to exercise a nominal request; it is not a live prototype. Modeled data flows, external provider integrations, and AI-service connections represent the target operational architecture and remain subject to future technical development, API provisioning, and task-based validation. Accordingly, the diagrams and fixed-data demonstration do not establish system effectiveness, improved investment decisions, user comprehension, implementation correctness, successful simulation, or production readiness; those outcomes require the validation activities defined later in the report.

2 Problem or Opportunity Statement

2.1 Problems and Opportunities

Beginner retail investors and students do not necessarily lack information; they often lack a clear way to turn that information into a fair comparison they can understand. The problem unfolds in the same order as a typical research task: first finding relevant evidence, then comparing companies on a common basis, and finally interpreting what the result means. The three areas below follow that journey and identify where StockLens could reduce the burden. The cited studies provide context for the needs examined in this report.

2.1.1 Fragmented Information

Relevant information is spread across many channels. FINRA reports that the most commonly used sources include brokerage research and tools (75 percent), recommendations from financial professionals (69 percent), business and finance articles (67 percent), and information from friends, family, or colleagues (65 percent) [1, p. 24]. Reliance also varies by user group: 85 percent of investors under 35 report using friends, family, or colleagues as an information source [1, p. 25]. When beginners compare stocks, they may therefore need to move between formal research, articles, market data, and informal advice. These sources can present information in different formats, cover different periods, and offer different levels of detail. Users must check that the information refers to the correct company and date, decide whether it is relevant, and organize it before making a meaningful comparison.

StockLens has an opportunity to reduce this information-gathering burden by organizing related records by company and date while keeping their sources visible. This would allow users to find and check the evidence for a comparison in one place. The intended improvement is connected and traceable information, rather than simply a larger collection of articles.

2.1.2 Inconsistent Comparison

After gathering information, beginners still need to decide how to compare companies. Market information does not arrive in one common unit: price momentum is expressed as a return over a period, volatility describes the size of price changes, and news describes events in words. Research shows that media tone can relate to market behavior, while financial text requires domain-specific treatment because common words can carry specialized meanings [2, 3]. Users must decide which factors matter, choose a time period for each factor, and determine how much importance to give each one. Without a defined method, these choices may change from stock to stock. A user might favor one company because its price rose during the last month, but favor another because of one positive headline, even though the two companies were judged using different evidence. The final preference may therefore reflect whichever fact was easiest to notice rather than a balanced comparison.

The problem is not that users have no information. It is that existing information does not provide a single basis for comparison. A price chart cannot be compared directly with a news article, and a low-risk stock is not automatically better or worse than a stock with strong momentum. Beginners must create their own method for combining these signals. Attention can also affect which evidence they emphasize. FINRA reports that 26 percent of investors use influencer recommendations at least sometimes; the percentage rises to 61 percent among respondents under 35 and 57 percent among investors with less than two years of experience [1, pp. 26–27]. Barber and Odean likewise found that individual investors were net buyers of attention-grabbing stocks [4]. If the comparison method is unclear or changes from stock to stock, the result becomes difficult to explain, repeat, or apply fairly.

StockLens has an opportunity to give users a common comparison framework. The same factors, time periods, calculation rules, and documented weights can be applied to every stock in the selected universe. Users could then see how each factor affected the result and compare companies on the same basis. This would make the process more consistent and explainable.

2.1.3 Difficult Interpretation

Much of the information available to investors assumes knowledge that beginners may not have. Financial websites and articles commonly use terms such as momentum, volatility, sentiment, market capitalization, and risk-adjusted return. A beginner may be able to read the value of an indicator without knowing what it measures, why it matters, or how it should affect a stock comparison. When several indicators point in different directions, the user must also decide how to interpret the conflict. This creates a gap between seeing financial information and understanding how to use it. This challenge is consistent with broader evidence that financial literacy matters for economic decisions and stock-market participation [5, 6].

FINRA provides evidence that this knowledge gap is present among investors. Across 11 investment knowledge questions, respondents answered an average of 5.3 correctly. Investors under 35 averaged 4.3 correct answers, and those with less than two years of experience averaged 4.0, compared with 6.0 among investors with at least ten years of experience [1, pp. 29–30]. These differences are directly relevant to StockLens because its intended users include beginners and students. If a platform only adds more charts and indicators, it may increase the amount of information without making the information easier to understand.

StockLens has an opportunity to translate technical information into clear language. It could define important indicators, explain why they matter, and show how positive and negative factors relate to a comparison. Users should be able to move from a technical value to a practical understanding of what that value contributes.

Together, these opportunities aim to help users find, compare, understand, and question stock information. They provide a basis for evaluating different approaches, without assuming that AI is necessarily required. The initial StockLens scope covers up to 100 predefined U.S. stocks, daily market data, and available news. Historical prices support learning about past changes; backtesting remains a possible future extension. Automated trading and personalized portfolio management are outside the initial scope. Interviews and task-based evaluation are still needed to confirm user priorities and test the proposed benefits.

3 Characterize the Solution Space

3.1 Stakeholders and Stakeholder Needs

StockLens is intended to help students and beginning investors compare stocks and understand the evidence behind a ranking. Four examples below show how people would use, review and support it. Their names and conversations are fictional; the supplied photos and avatars are visual references, not records of interviewed participants. The needs still require stakeholder review.

3.1.1 Primary Research Users

User A: Maya Chen — Student researcher



Maya is preparing a two-stock comparison for a student investment discussion. She can read a price chart, but wants to explain why the stocks rank differently. She worries that she will repeat an AI explanation without understanding it. She needs a consistent comparison, an explanation she can follow and sources she can inspect (ST-01; N-01 to N-03).

Maya: “Why is Stock A above Stock B?”

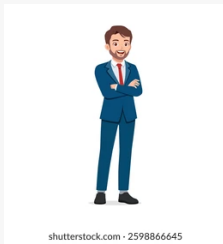
Proposed response: Show the score components and explain the two main drivers using those inputs.

Maya: “Can I open the article behind that claim?”

Proposed response: Show the supporting source, its publication date and the market-data date.

Figure 1: User A: an illustrative student research scenario using a user-provided reference photo.

User B: Daniel Brooks — Beginning investor



Daniel has a short list of companies to research after work. He wants to narrow that list without treating a high rank as an instruction to buy. His main concern is comparing fresh information for one stock with old information for another. Like Maya, he needs consistent scores and inspectable evidence, with particular attention to dates and the limits of a ranking (ST-01; N-01 to N-03).

Daniel: “Are both scores based on the same date?”

Proposed response: Show the analysis date and input coverage so he can judge whether the comparison is fair.

Daniel: “Does first place mean this is right for me?”

Proposed response: Explain that the rank summarizes selected indicators. Personal suitability and trade execution are outside StockLens’s scope.

Figure 2: User B: an illustrative independent research scenario using a user-provided avatar.

3.1.2 Supporting and Life Cycle Roles

Maya and Daniel perform the main research use case, UC.1. Elena and Ben represent life cycle roles. Their review and maintenance tasks help define stakeholder needs without adding new participants to the normal research sequence.

User C: Elena Park — Project reviewer	
	Elena reviews the course submission. She needs to follow a claim in the report back to a need, a model element and the evidence from the prototype. She looks for a clear link between each need, its model actions and the planned acceptance evidence (ST-03; N-05). Elena: “Which user need does this action address?” Proposed response: Show the need ID, the relevant use-case action and the matching model entity. Elena: “Was this explanation tested with users, or is it a fixed demo response?” Proposed response: Identify the evidence type and report the actual validation status, including work that has not been done.

Figure 3: User C: an illustrative review role using a user-provided avatar; Elena is not a named course evaluator.

User D: Ben Carter — Student developer and maintainer	
	Ben takes over the demonstration from a teammate. He needs to start it in the agreed environment and know which configuration produced a result. Later, he may need to change a provider or close the service when the project ends. His development and maintenance responsibilities span ST-02 and ST-04, including resource feasibility and repeatable startup and shutdown (N-04, N-06). Ben: “Can I run this without asking the original developer?” Proposed response: Provide the environment, launch steps and configuration needed to repeat the demonstration. Ben: “What must I hand over or shut down?” Proposed response: Identify ownership, permitted records, service credentials and subscriptions in a transition checklist.

Figure 4: User D: an illustrative student team role. The supplied cartoon is a visual label, not an age or demographic claim.

These four examples cover the identified human roles. Market/news providers and the external AI provider remain separate stakeholders (ST-05 and ST-06); their needs concern service use and permitted inputs, not a dashboard session.

3.1.3 Stakeholder Roles, Concerns, and Inclusion

The primary research user is represented by ST-01. ST-02 to ST-04 cover acquisition through retirement, while ST-05 and ST-06 supply external capabilities. Although project members may initially fill several roles, their responsibilities are kept distinct so that delivery, evaluation and maintenance needs are not overlooked.

Table 1: Preliminary stakeholder profiles and the reason for including each group.

Stakeholder / role	Needs and concerns	Why included
ST-01 Students and beginning investors Use and feedback	Consistent comparisons (N-01), understandable ranking drivers (N-02), and inspectable sources and dates (N-03). Concern: unexplained scores, stale evidence and misplaced confidence in AI text.	They perform UC.1 and determine whether the research workflow is useful. Their feedback should guide explanation detail and presentation.
ST-02 Project team Acquisition, development, deployment	A feasible scope within available time, skills and service resources (N-04). Concern: growing integration effort, uncertain access and unconfirmed costs.	The team selects providers, builds the prototype and resolves trade-offs between user benefit and delivery effort.
ST-03 Instructor and evaluators Review and acceptance	Traceable, consistent evidence linking the problem, needs, model and prototype (N-05). Concern: unsupported claims or diagrams that disagree with the implementation.	They assess the course deliverables and need to distinguish a connected demonstration from validated user benefit.
ST-04 System maintainer Support and retirement	Repeatable operation and documented handover or shutdown (N-06). Concern: undocumented configuration, provider changes and abandoned credentials.	Someone must be able to sustain or retire the system. The project team fills this role provisionally; no long-term operator is designated.
ST-05 Market data and news providers External service provision	Use of their services and content within agreed access, retention and display conditions (N-07). Concern: unsupported redistribution or requests outside selected access limits.	The concept depends on their records, timestamps and source identifiers; their conditions constrain the feasible design.
ST-06 External AI service provider External analysis service	Intended inputs through approved service access (N-08). Concern: inappropriate payloads or use outside selected service conditions.	The current boundary places model hosting outside StockLens. This stakeholder must be revisited if that architectural assumption changes.

Relation to the model. ST-01 corresponds to X.01 Research User; ST-04 to X.05 System Maintainer; ST-05 covers X.02 Market Data Provider and X.03 News Provider; ST-06 corresponds to X.04 AI Model Service. ST-02 and ST-03 are life cycle stakeholders rather than additional participants in the normal research sequence.

Trade-offs. More news coverage and AI calls can improve context while increasing cost and review effort. Short explanations are easier to scan but may hide uncertainty. The initial daily-data scope and limit of 100 US stocks bound the workload; users should still be able to open the underlying evidence rather than rely solely on the short explanation.

3.1.4 User Needs and Measures of Effectiveness

N-01 to N-03 express the intended user outcomes. The proposed measures assess whether users can complete a meaningful research task and explain its result.

Table 2: User needs, preliminary MOEs and proposed validation criteria.

Need / owner	MOE and proposed criterion	Validation method and evidence
N-01 ST-01: Compare stocks on a consistent basis.	Comparison success rate: participants who complete the defined comparison without assistance / all participants attempting it. Proposed target: $\geq 80\%$.	Ask each participant to compare two stocks on the same analysis date using two agreed indicators and the displayed ranking. Score against a prepared task rubric; retain task answers, completion and assistance records.
N-02 ST-01: Understand the main drivers of a displayed ranking.	Driver comprehension rate: participants correctly identifying both requested drivers / all participants attempting the task. Proposed target: $\geq 80\%$. Guardrail: no contradiction of computed indicators in the reviewed explanations.	Ask why Stock A ranks above Stock B. Compare the answer with a checked scoring breakdown, using the documented factor values and weights. Audit the explanation against actual inputs and record each discrepancy.
N-03 ST-01: Inspect the evidence and judge its freshness.	Evidence-location success rate: participants locating the requested source and date / all participants attempting the task. Proposed target: $\geq 80\%$. Guardrail: every sampled news claim has an identifiable, supporting source.	Ask the participant to open the source of a displayed news claim and identify its publication date and the market-data date. Separately review all news claims in the selected explanations for source identity and actual support.

3.1.4.1 Proposed pilot procedure

1. Prepare. Recruit participants from the intended student/beginner group. Record relevant experience, agree the sample size, and freeze the task instructions and scoring rubric before testing. Use a checked two-stock case with differing score drivers and a source with an identifiable publication date. Confirm the proposed thresholds with the team.

2. Observe. Give each participant the same task instructions and access to the same dated dataset and scoring configuration. Record success, errors and requests for help. A task requiring facilitator guidance counts as assisted, not an unassisted success.

3. Check. Audit explanations against the underlying score components and cited material. Check that each cited passage supports the associated claim.

4. Report. Present counts as well as percentages, sample characteristics and limitations. Report each MOE separately; a high comparison score must not conceal poor explanation grounding. Compare N-01 task completion and time with the same research task using the separate source pages; treat any time improvement as exploratory until the team agrees a target.

Interpretation. Confirm the preliminary 80% thresholds before the pilot. Use representative content and observed user tasks, and interpret the findings within the recruited sample.

3.1.5 Supporting Needs and Acceptance Evidence

N-04 to N-08 support delivery and continued operation. Their measures concern feasibility, traceability, maintainability and controlled external-service use. They complement the user MOEs.

Table 3: Supporting stakeholder needs, preliminary MOEs and proposed acceptance criteria.

Need / owner	Preliminary measure and criterion	Method and retained evidence
N-04 / ST-02 Build a manageable system within available time, skills and service resources.	Core demonstration completion. The agreed UC.1 demonstration runs in the approved environment within confirmed time and resource limits.	Demonstrate the agreed path; review the environment, selected service plan and resource record. Document the actual limits before judging acceptance.
N-05 / ST-03 Trace the problem, needs, models and prototype to consistent evidence.	Traceability coverage. 8/8 needs have a validation method; every participant in the selected skeleton maps to an identified model entity.	Inspect the need-to-validation register and the use-case/model/prototype mapping. Retain entity IDs, participant names and unresolved inconsistencies. Recalculate coverage if the baseline changes.
N-06 / ST-04 Maintain repeatable operation and documented handover or shutdown.	Independent task completion. A second team member completes startup and shutdown without undocumented steps. A transition checklist is recorded.	Have a member other than the instruction author follow the documented process. Record missing steps and repeat after correction. Review ownership, configuration, retained records and credential revocation for handover/retirement.
N-07 / ST-05 Use provider services and content within agreed conditions.	Provider review coverage. Every selected market/news provider has documented access, retention and display conditions, with intended use checked against them.	Inspect selected provider documentation and the planned requests, storage and display. Retain dated references and decisions. Do not assume a source permits redistribution because it is publicly accessible.
N-08 / ST-06 Send only intended inputs through approved AI service access.	Payload conformance. Every inspected request matches the approved field list, with no credentials or unnecessary personal data in the analysis content.	Inspect the approved input schema and representative explanation requests. Retain redacted examples and review results. Service authentication is handled separately from the content supplied for AI analysis.

3.1.5.1 From stakeholder to validation evidence

Stakeholder → Need → Measure → Validation evidence

Example: ST-01 → N-02 → driver comprehension → scored participant answers plus an explanation-to-input audit. Assumptions A-01, A-02 and A-04 identify the data and AI dependencies that must also hold.

Retain a review record for each need: owner, method, threshold, evidence location and status. All needs are currently *not yet validated*; revise them and their criteria as stakeholder feedback and implementation evidence become available.

3.1.6 Operational Situations to Be Addressed

The normal research path is only one part of the stakeholder picture. Table 4 records the main variations and support situations identified so far. It is a review checklist, not an exhaustive set of possible failures. Rows outside UC.1 identify later exception and support scenarios.

Table 4: Scenario coverage and the needs to revisit in each situation.

Situation	User concern and proposed treatment	Coverage / needs
S-01 Normal comparison Maya or Daniel compares two eligible stocks.	Use a common analysis date and explain the displayed score components. The user can open the cited evidence and compare another stock.	UC.1; N-01 to N-03.
S-02 Stale or missing data One stock lacks a required observation.	Make the date and coverage gap visible. Do not present an incomplete score as directly comparable to a complete one. Agree the eligibility or hold rule before adding this exception flow.	Later exception; N-01, N-03; A-01, A-04.
S-03 No usable news No relevant article is available for a stock.	Distinguish “no usable evidence” from neutral sentiment. If the scoring configuration requires that input, hold the affected result until a missing-input rule is agreed.	Later exception; N-02, N-03; A-04.
S-04 Conflicting AI explanation Text disagrees with a score or source.	Check the claim against the calculation and cited content. A reviewer should record the discrepancy; a failed check must not be counted as a valid explanation. Define correction/display behavior in later design.	Explanation audit; N-02, N-03, N-05.
S-05 Service or quota failure A data, news or AI call cannot complete.	Identify which result is unavailable. Distinguish unavailable source data from an unavailable AI explanation; do not present an unexplained or incomplete result as the complete research output. Recovery, retries and cached-result rules need separate design.	Later exception; N-04, N-06; A-01, A-02, A-05.
S-06 Unsupported request The user selects an ineligible stock or asks to trade.	Explain the supported universe and daily research scope. Trading, personal portfolio advice and intraday support remain outside the boundary. Invalid-input handling is beyond the nominal precondition.	Scope review; N-01, N-04; A-05.
S-07 Review of a demo Elena asks what was demonstrated.	Trace the observed call path to model entities. Separate fixed outputs and planned tests from completed user validation; retain the actual evidence and unresolved gaps.	Review; N-05.
S-08 Configuration change Ben changes a score rule, prompt or provider.	Record the changed configuration and recheck input alignment and explanations. Repeat affected validation activities before claiming that earlier results still apply.	Support; N-02, N-04, N-06.
S-09 Access or input restriction Provider terms or approved fields change.	Review requests, retained content and display. Check AI payloads against the allowed fields; keep authentication separate from analysis content. Revisit the provider choice if necessary.	External services; N-07, N-08.
S-10 Handover or retirement The team transfers or stops the service.	Use the documented transition process: name the owner, retain only permitted records and revoke unneeded access. Have another member check that the process can be followed.	Life cycle; N-06 to N-08.

The pilot in Table 2 evaluates normal research. The other situations define the next scenario-validation tasks.

3.2 Operational Concept (OpsCon)

This operational concept follows a StockLens comparison from the user's request to the displayed result. It identifies what happens inside the system, which services and people remain outside it, and what information crosses the boundary. Reading the Asset Diagram as both an Operational Concept (OpsCon) and a System Context Diagram (SCD/SOI) keeps the narrative and the architecture aligned. The separate Universe Hierarchy then shows how the StockLens System, its internal assets, and its external participants fit together.

3.2.1 Scope, System of Interest, and Boundary

The system of interest is the StockLens System, an AI-assisted research and ranking platform. Its job is to help a user compare and understand stocks, not to make or execute an investment decision. The initial scope covers up to 100 U.S. stocks, daily market observations, company information, relevant news, comparative rankings, historical price displays, and explanations for individual stocks. Automated trading, order execution, personalized portfolio management, and intraday support remain outside the SOI boundary.

The Universe hierarchy reproduced in this report decomposes StockLens into four internal assets: 1.1 Scoring & Ranking Model, 1.2 Historical Validation Module, 1.3 AI Explanation Module, and 1.4 Dashboard / User Interface. The external environment contains X.01 Research User, X.02 Market Data Provider, X.03 News Provider, X.04 AI Model Service, and X.05 System Maintainer. These external assets participate in the operating environment but are not components of the SOI. The boundary separates the functions developed and configured by the StockLens team from users and independently operated providers. This makes data availability and AI-service behavior explicit external dependencies. The Historical Validation Module is a provisional model asset; historical-price display is in scope, while performance backtesting remains a potential extension rather than an implemented capability.

3.2.2 OpsCon and SCD/SOI Context

Figure 5 serves two complementary purposes. From the OpsCon perspective, it shows the environment in which StockLens performs its research mission and identifies the external actors and enabling systems required for normal operation. From the SCD/SOI perspective, StockLens System is the SOI and the X-prefixed assets remain outside the system boundary. The enclosing rectangle identifies the StockLens SOI; Research User and X.02–X.05 remain outside it. The five named interfaces cross the boundary. The rectangle is a graphical annotation saved in the Innoslate context view, not an additional system component. This boundary places scoring, orchestration and presentation under team control while treating source availability and external model behavior as dependencies.

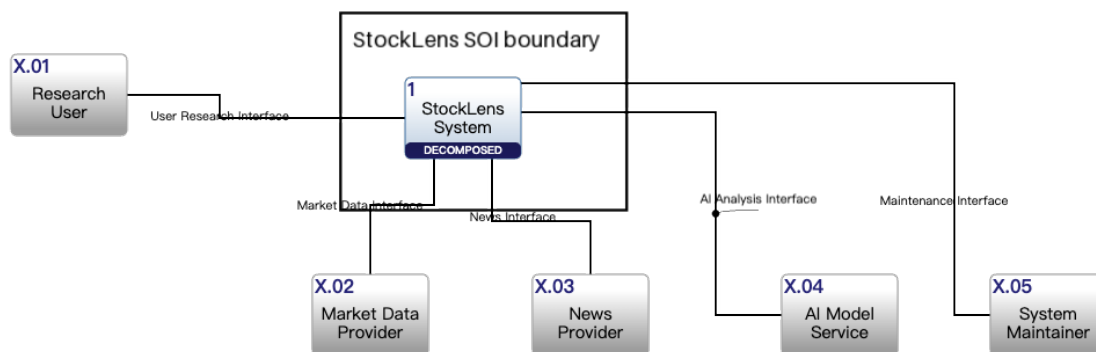


Figure 5: StockLens Universe Context Asset Diagram used for the OpsCon and SCD/SOI perspectives.

3.2.3 External Interfaces

Table 5 summarizes the major boundary exchanges shown by the context model. The interface names match the Innoslate Asset Diagram, while the exchanged information captures the current report-level definition of what enters and leaves StockLens.

Table 5: External interfaces for the StockLens System.

External entity	Interface	Into StockLens	Out of StockLens
X.01 Research User	User Research Interface	Stock pair, analysis date, and research request	Rankings, comparisons, explanations, and sources
X.02 Market Data Provider	Market Data Interface	Daily prices, volume, and company records	Symbol, date-range, and field requests
X.03 News Provider	News Interface	Articles or permitted text, source identifiers, and timestamps	Company, keyword, and date-range queries
X.04 AI Model Service	AI Analysis Interface	Evidence-linked explanation text	Computed score breakdown, driver values, relevant news, dates and source identifiers
X.05 System Maintainer	Maintenance Interface	Approved configuration and update actions	Service status and maintenance information in the future complete system

3.2.4 Operational Concept Narrative

A normal session begins when a Research User chooses eligible stocks and an analysis date in the Dashboard/User Interface. StockLens then requests market and company records through the Market Data Interface and dated news evidence through the News Interface. The Scoring & Ranking Model applies the same documented factors, observation periods, and weights to both stocks, so the comparison is created before any explanation is written. Only after the score breakdown is complete does the AI Explanation Module send the relevant results and evidence to the external AI Model Service. The returned text explains the calculation; it cannot change the scoring rules or independently decide the ranking. A sentiment factor would enter this flow only after its derivation has been separately specified and checked.

The Dashboard/User Interface brings the result back to the user as a connected story: what was compared, which factors mattered, what evidence supports the explanation, and when that evidence was dated. The user can inspect the sources or begin another comparison. The Maintenance Interface supports approved configuration, updates, and future service-status exchanges, but it does not participate in the normal research path. The session ends with information for the user to review; no trade is placed and no personalized portfolio is created.

3.2.5 Universe Hierarchy

Figure 6 shows the Universe as the highest-level asset. StockLens System and the five external assets are all directly contained within the Universe, making clear that the Research User, providers, AI service, and maintainer are part of the operating environment rather than internal StockLens components. StockLens is then decomposed into its four current internal model assets.

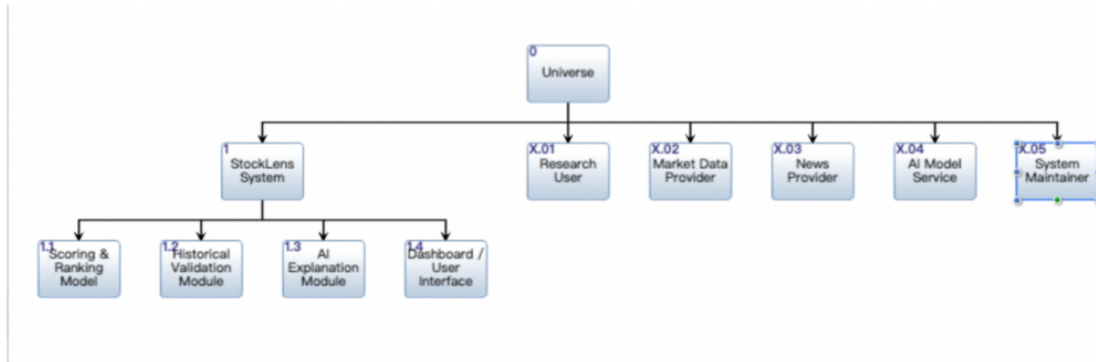


Figure 6: StockLens Universe Hierarchy showing the SOI, external assets, and current internal decomposition.

3.2.6 Model Consistency and Transition

The Universe Context Asset Diagram and Universe Hierarchy display matching StockLens and X.01–X.05 asset labels. The context view emphasizes the SOI boundary and external interactions, while the hierarchy view emphasizes where each asset belongs. These labels define the report’s reference asset structure. The revised behavioral model reuses X.02 Market Data Provider, X.03 News Provider and X.04 AI Model Service as separate performing assets. UC.1.4.1 returns market records, UC.1.4.2 returns dated news evidence, and UC.1.6 explains the score-evidence package produced by StockLens. Dashboard/UI is an internal presentation role: an explanation delivered to this consumer remains an exchange into the StockLens SOI. These are logical responsibilities, not a specification of direct browser-to-provider network calls. Universe entity 199040 and behavior parent 185145 define the report baseline; other hierarchy drafts are not used as alternative baselines.

3.2.7 Primary Use Case

3.2.7.1 Use Case Characterization

Goal: Enable a student or retail investor to compare two eligible U.S. equities on a common date, inspect normalized score drivers, and review supporting news evidence without misinterpreting ranking outputs.

Primary Actor: ST-01 Research User (Students and Beginning Investors).

Preconditions:

- System scope constraints hold: selected equities belong to the predefined universe (≤ 100 U.S. stocks).
- Assumptions A-01, A-02, A-04 and A-05 hold: the required sources and AI service are accessible, the inputs are aligned to the requested analysis date, and a documented scoring configuration is available.

Trigger: The user selects two target stock tickers and initiates a comparison query on the dashboard.

Completion Conditions: The interface displays a synchronized comparison view containing normalized score breakdowns, overall relative ranks, plain-language driver explanations, and auditable source citations.

3.2.7.2 Use Case Diagram and Interpretation

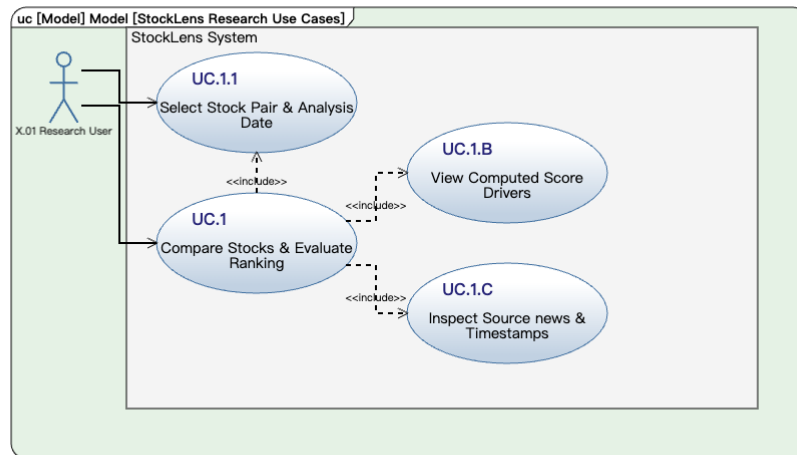


Figure 7: Use Case Diagram for the StockLens Primary Research Workflow (UC.1).

Figure 7 depicts the boundary between the Research User and the StockLens System. The primary actor interacts with the core operational capability, *UC.1: Compare Stocks & Evaluate Ranking*. To support transparent and repeatable comparisons, the use-case view identifies three included capabilities. The inclusion links express required functionality, not execution order:

- **Select Stock Pair & Analysis Date:** Defines the two stocks and common analysis date. This supports N-01, but input-date and coverage checks are still required before a comparison can be accepted.
- **View Computed Score Drivers:** Shows the factor values and contributions behind the aggregate score. This supports N-02; user comprehension must be evaluated separately.
- **Inspect Source News & Timestamps:** Lets the user inspect the source and date of a displayed news claim. This supports N-03; the source audit must establish that the cited material actually supports the claim.

3.2.8 Normal Operating Flow

3.2.8.1 Normal Operating Sequence

The intended UC.1 scenario contains nine actions allocated across six performing assets. The following narrative describes their data-dependent order:

1. **UC.1.1 Select Stock Pair & Analysis Date (Research User):** The user accesses the interface, selects two target equity tickers from the predefined universe of up to 100 stocks, and specifies a common historical or current trading date.
2. **UC.1.2 Transmit Evaluation Request (Dashboard / UI):** The interface sends the selected stock pair and analysis date to StockLens as an evaluation request. The transport and request schema remain implementation decisions.

3. **UC.1.3 Query External Data Feeds (StockLens System):** StockLens uses the request to obtain the required market and news records from the selected external providers. Caching is not established by this model.
4. **UC.1.4.1 Return Market Data (X.02 Market Data Provider):** The provider returns market records and observation dates for the requested pair.
5. **UC.1.4.2 Return News Evidence (X.03 News Provider):** The news provider receives the same data-fetch request and returns relevant news with source identifiers and publication dates. Both provider outputs feed UC.1.5.
6. **UC.1.5 Calculate and Normalize Ranking (StockLens System):** StockLens computes a comparison using a common factor definition, observation period, normalization method and weighting configuration. Momentum and volatility are candidate factors. Exact formulas, scales and weights require a documented baseline; a sentiment factor requires its own input and derivation specification.
7. **UC.1.6 Synthesize Grounded AI Explanation (X.04 AI Model Service):** The external AI service is intended to receive the computed score drivers and linked news evidence and return a plain-language explanation. The shared I/O object now carries the score breakdown, driver values, relevant news evidence, source identifiers, publication dates and requested analysis date. The service explains these supplied results and must not change their ranking. Source checks and time-alignment checks are still needed to detect unsupported statements or look-ahead bias.
8. **UC.1.7 Render Comparison Dashboard (Dashboard / UI):** The dashboard ingests both the quantitative score breakdowns and the AI narrative, updating visual comparison tables, charts, and auditable citation links.
9. **UC.1.8 Inspect Score Drivers and Citations (Research User):** The user reviews the comparison, opens the cited sources and examines the dates and ranking drivers.

3.2.8.2 Action Diagram and Interpretation

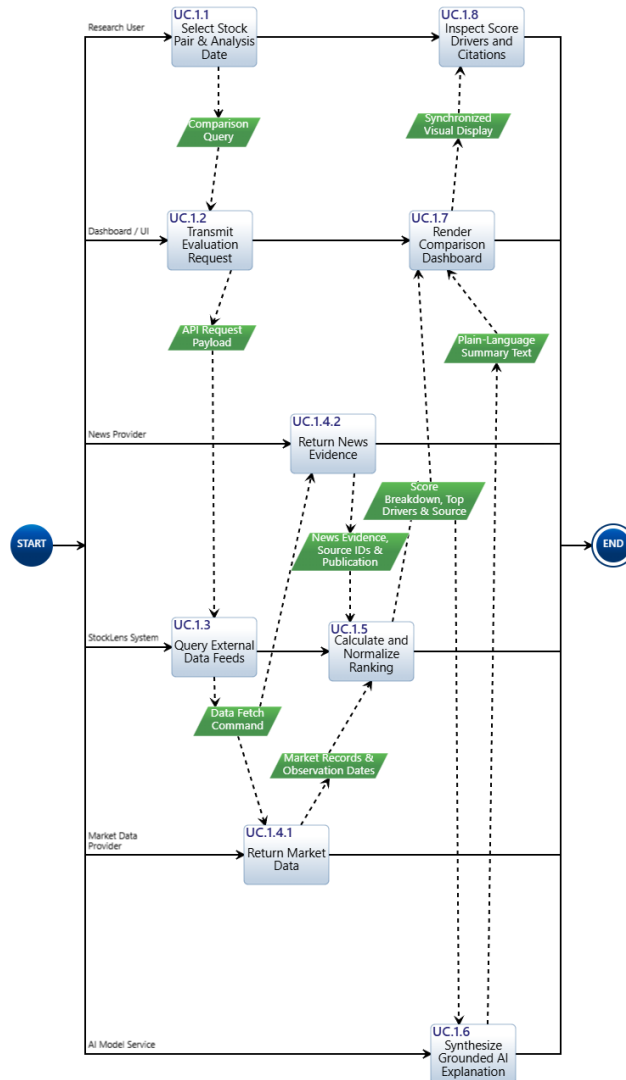


Figure 8: StockLens Action Diagram for UC.1 Data Processing and Delivery Flow.

Figure 8 illustrates the end-to-end transformation of inputs and outputs through the StockLens functional architecture. Six performing assets separate responsibilities: Research User, Dashboard/UI, StockLens System, X.02 Market Data Provider, X.03 News Provider and X.04 AI Model Service. Market records and news evidence arrive through separate provider actions. StockLens owns scoring and packages the evidence supplied to the AI service.

The named data dependencies explain the intended ordering: market and news providers (UC.1.4.1 and UC.1.4.2) must return their dated records before quantitative scoring (UC.1.5) executes. The AI explanation action (UC.1.6) cannot execute independently from raw news alone; it requires the structured *Score Breakdown, Top Drivers & Source Evidence* emitted by UC.1.5. This dependency is intended to anchor explanations to calculated metrics and support N-02. Explanation audits must

still confirm that the generated text matches its inputs; provider limits must be checked separately.

3.2.8.3 Action Input and Output Specification

The score-evidence package contains the stock pair, analysis date, scores, factor contributions, relevant news excerpts, source identifiers and publication dates. UC.1.5 preserves source evidence received from UC.1.4.1 and UC.1.4.2; it does not generate new source claims. UC.1.6 returns explanation text linked to those same identifiers. This defines the payload needed for N-03 and keeps explanation generation separate from scoring.

Table 6 summarizes the functional decomposition, defining the specific performing assets, trigger inputs, and generated outputs across the proposed operational actions in UC.1. The final row records a user outcome rather than a modeled machine output; the visible Action diagram does not show an outgoing I/O object for that inspection.

Table 6: StockLens Action and Input/Output (I/O) Specification.

Action Entity	Performing Asset	Input(s)	Output(s)
UC.1.1 Select Stock Pair & Analysis Date	Research User	User Selection / Target Date	Comparison Query
UC.1.2 Transmit Evaluation Request	Dashboard / UI	Comparison Query	API Request Payload
UC.1.3 Query External Data Feeds	StockLens System	API Request Payload	Data Fetch Command
UC.1.4.1 Return Market Data	X.02 Market Data Provider	Data Fetch Command	Market Records & Observation Dates
UC.1.4.2 Return News Evidence	X.03 News Provider	Data Fetch Command	News Evidence, Source IDs & Publication Dates
UC.1.5 Calculate and Normalize Ranking	StockLens System	Market Records & Observation Dates; News Evidence, Source IDs & Publication Dates	Score Breakdown, Top Drivers & Source Evidence
UC.1.6 Synthesize Grounded AI Explanation	X.04 AI Model Service	Score Breakdown, Top Drivers & Source Evidence	Plain-Language Summary Text
UC.1.7 Render Comparison Dashboard	Dashboard / UI	Score Breakdown, Top Drivers & Source Evidence, Plain-Language Summary Text	Synchronized Visual Display
UC.1.8 Inspect Score Drivers and Citations	Research User	Synchronized Visual Display	Research inspection completed (user outcome)

3.2.9 Activity Diagram and Interpretation

Figure 9 presents the nine actions as a connected research activity. The user selects a stock pair and date (UC.1.1), the interface transmits the request (UC.1.2), and the system obtains market and news records (UC.1.3, UC.1.4.1 and UC.1.4.2). Ranking calculations (UC.1.5) supply the score breakdown used for explanation generation (UC.1.6). The dashboard renders the comparison (UC.1.7), after which the user inspects the drivers and citations (UC.1.8). The named information flows explain what must be available before the next result can be produced.

The diagram covers the nominal path. Provider failures, invalid inputs and recovery paths belong to separate exception scenarios.

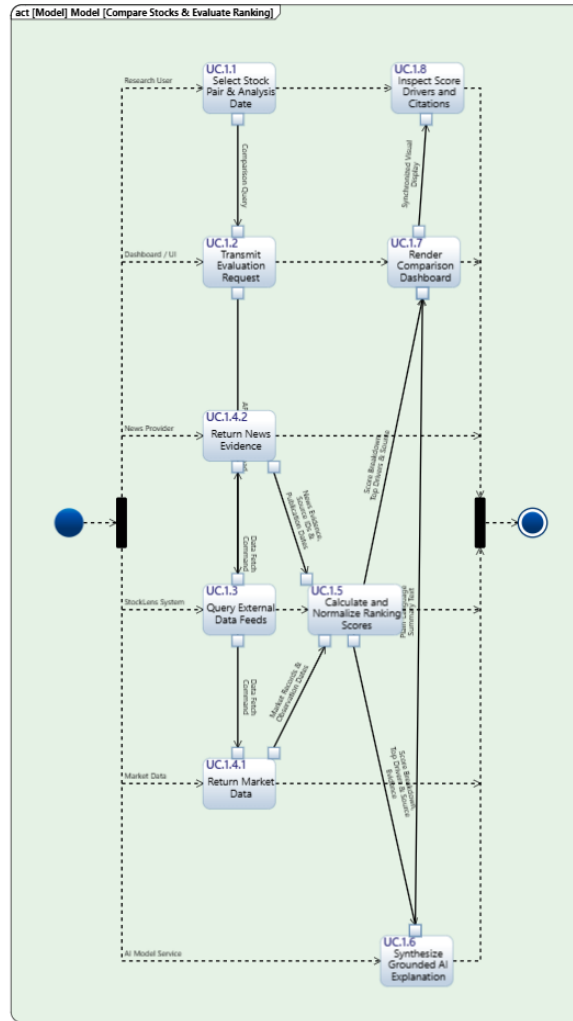


Figure 9: Activity diagram for the StockLens comparison workflow.

3.2.10 Sequence Diagram and Interpretation

Figure 10 orders the interactions between Research User, Dashboard/UI, StockLens System and the three provider assets from top to bottom. The request passes from the user to the dashboard and system; external records return before the ranking and explanation stages. UC.1.7 delivers the comparison to the Research User, who then performs UC.1.8 in the Action and Activity views. Inspection is a terminal human activity, with no subsequent service request.

UC.1.5 is performed by StockLens System, which supplies the calculated score-evidence package to X.04 for explanation generation. X.04 returns the explanation to the Dashboard/UI, the functional consumer within the StockLens boundary. Market records and news evidence are returned separately by X.02 and X.03. The shared Data Fetch Command supplies both provider actions, as specified in Table 6. Provider access is to be mediated by the system interface.

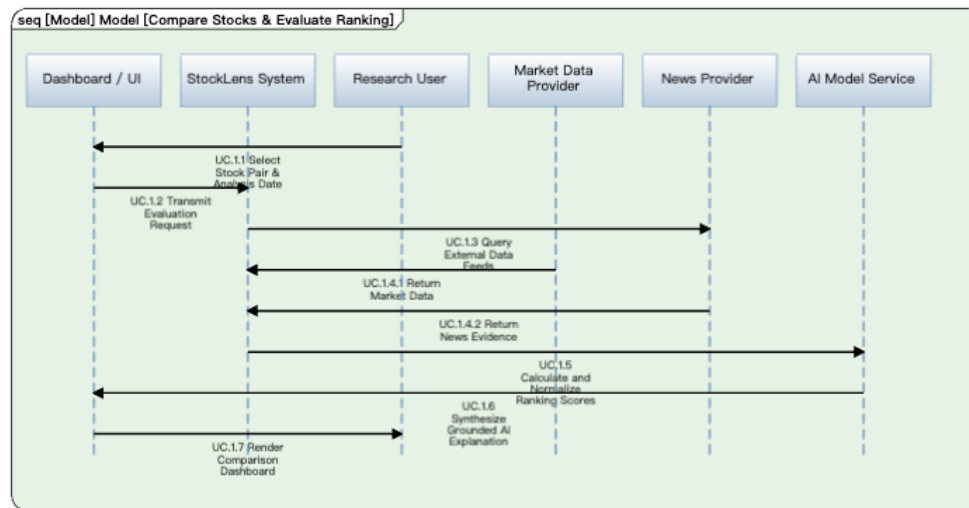


Figure 10: Sequence diagram for the StockLens comparison and explanation workflow.

3.2.11 Spider Diagram and Action Allocation

Figure 11 links the parent UC.1 action to its nine child actions and their performing assets. Research User performs selection and inspection; Dashboard/UI performs request transmission and rendering; StockLens System performs retrieval coordination and ranking; X.02 returns market data, X.03 returns news evidence, and X.04 generates the grounded explanation. The Spider view records decomposition and responsibility, not message timing.

The revised Action model reuses the context assets X.02, X.03 and X.04 rather than a combined External API asset. The maintainer remains outside the normal research flow.

Table 7: Platform goals, model responsibilities and evidence needed.

Goal / need	Model contribution	Observable result and check
Consistent comparison (N-01)	Selection defines a stock pair and date; retrieval supplies common inputs; UC.1.5 assigns ranking to StockLens.	A comparison shows both stocks, analysis date and score components. Check identical factor definitions and input periods, then run the N-01 user task.
Understand the ranking (N-02)	UC.1.5 provides the score breakdown; UC.1.6 uses it to produce an explanation; UC.1.7 presents the result.	The explanation identifies the actual drivers without changing the numerical ranking. Audit agreement with calculated inputs and score the N-02 comprehension task.
Inspect the evidence (N-03)	X.02/X.03 supply records; UC.1.7 displays the result; UC.1.8 assigns inspection to the user.	The user can locate the cited source and relevant dates. The score-evidence I/O now specifies source IDs, dates and supporting content; check source support and perform the N-03 task.
Build and review a feasible prototype (N-04–N-06)	Context defines external dependencies; behavioral views identify the request path and responsibilities.	A separate Walking Skeleton should pass one request through named participant stubs and return fixed, correctly shaped output. Retain the call list and demonstration evidence, then evaluate provider data and AI quality through the planned audits.

For example, Maya selects two eligible stocks on the same date. The proposed platform obtains their records, calculates the documented comparison, and passes the score breakdown to the explanation service. Maya then sees the scores and reasons together and opens a supporting source. The model connects these steps to N-01–N-03; a future demonstration must show that the steps are wired together, and the pilot must establish whether she can complete and understand the task.

The Lab Manual’s Walking Skeleton is narrower than this full concept: one end-to-end request uses hard-coded data and model stubs, with no real provider or AI calls. The local fixed-stub implementation exercises a nominal request. Code paths, model entity IDs and demonstration results should be linked in the GitHub documentation.

3.2.13 Cross-View Consistency Review

The report exports were compared with the corresponding live views in Innoslate project 659. The Action, Activity, Sequence, and Spider views now use the same UC.1 parent entity (185145), action identifiers, provider assets, performing-asset allocations, and evidence flows. The four views were re-exported after these elements were reconciled in the shared model.

Table 8: Cross-view consistency review of the harmonized UC.1 model.

Check	Harmonized state	Verification in report baseline
Action coverage	UC.1.1 identifies stock-pair and date selection, and UC.1.8 identifies the user's inspection of score drivers and citations.	Both actions appear consistently in the Action and Activity views; the Sequence view delivers the completed comparison to the Research User.
Calculation ownership	UC.1.5 is performed by the StockLens System and produces the score breakdown, top drivers, and source-evidence package.	The Action allocation, I/O specification, Sequence exchange, and Spider allocation all retain StockLens ownership.
Explanation routing	X.04 receives the score-evidence package and returns plain-language summary text to the Dashboard/UI.	The same sender, receiver, and payload roles are used in the Action, Sequence, and I/O views, with provider access mediated by StockLens.
External assets	UC.1.4.1/X.02 returns market data, UC.1.4.2/X.03 returns news evidence, and UC.1.6/X.04 generates the explanation.	The Action, Sequence, and Spider exports reuse the same three provider entities and keep market and news inputs distinct.
Identifiers and baseline	Universe 199040 and UC.1 parent 185145 define the report baseline; UC.1.4.1 and UC.1.4.2 identify the two provider responses.	Captions, narrative descriptions, tables, and exported views use the same identifiers and hierarchy.
Evidence and scope	The nominal workflow ends with source and driver inspection rather than trade execution.	The validation table retains explanation audits, source-support checks, time-alignment checks, and user-task measures; exception flows are reserved for separate models.

3.3 Preliminary Life Cycle Concepts

The following concepts describe the proposed responsibilities and transition conditions across the StockLens life cycle. This conception-through-retirement view follows the system life-cycle process perspective established in ISO/IEC/IEEE 15288 [7].

3.3.1 Acquisition Concept

The project team should select market, news and AI services against the bounded daily research task, rather than maximize the number of integrations. Before committing to a source, the team should confirm coverage, timestamps, permitted retention and display, access limits and resource requirements. A small representative dataset and a rule-based comparison provide the initial evaluation baseline. Acquisition is complete when the selected sources can support that baseline within the team's confirmed resources and the N-07/N-08 reviews are recorded.

3.3.2 Deployment Concept

Deployment should begin with a controlled demonstration using a fixed analysis date, documented inputs and a versioned scoring configuration. The team should then check the complete UC.1 path in the intended browser environment before inviting pilot participants. Credentials remain outside displayed content and shared examples. The deployment record should identify the software version, configuration, provider access and known limitations so that a second team member can repeat the demonstration.

3.3.3 Support Concept

A designated team member should maintain provider access, configuration records and instructions for startup, routine checks and shutdown. Missing or stale data, quota failures and explanation discrepancies should be recorded with the affected analysis date and component. Changes to a source, score rule or explanation prompt require the affected checks to be repeated. Handover is assessed through N-06: another member follows the instructions without undocumented assistance. A permanent operator beyond the course has not yet been designated.

3.3.4 Retirement Concept

When the course demonstration ends or ownership changes, the team should decide whether to transfer or close the service. It should preserve the report, model, configuration and evidence that it is permitted to retain, remove records that may no longer be held, revoke unnecessary credentials and stop unused subscriptions. The outgoing and incoming owners should record any unresolved limitations. Retirement is complete when the service no longer depends on unattended access and the retained project records can be interpreted without the original developer.

3.4 Alternative Solution Classes and Evaluation

An alternative solution class is a broad type of product that could address the user problems in Section 2. This part compares four product-level approaches: an information aggregation dashboard, a rule-based stock screening and ranking platform, an AI-assisted research and ranking platform, and an automated investment or trading system.

The comparison focuses on how each approach could support a repeatable and understandable stock comparison. Each alternative is evaluated against consistent comparison (N-01), understanding ranking drivers (N-02), and inspecting sources and dates (N-03), together with delivery feasibility (N-04). The analysis also considers the system boundary and the practical effort required for a course project. The alternatives are compared qualitatively because stakeholder priorities and development effort have not yet been measured well enough to support a weighted scoring method.

3.4.1 Information Aggregation Dashboard

An information aggregation dashboard brings market data, company information, charts, and financial news into one interface. It can organize records by company and date and provide links to the original sources. The user can view the same types of information for several stocks without repeatedly moving between unrelated websites. However, the user still decides how to compare the stocks and how to interpret the evidence.

Bringing the sources together is expected to improve access and reduce repeated information gathering; the amount of effort saved still needs to be observed. A shared page layout can also make basic side-by-side review easier. The main limitation is the absence of a common scoring method or an explanation of why one stock compares more favorably than another. Users may still focus on different indicators, use different time periods, or reach conclusions without understanding conflicting evidence. The dashboard is therefore useful, but it addresses only part of the identified need.

3.4.2 Rule-Based Stock Screening and Ranking Platform

A rule-based platform adds a documented comparison method to integrated data. The system applies the same factors, time periods, weights, and calculations to every stock, then presents scores, rankings, charts, factor values, and source links. Fixed rules or templates can explain direct relationships, such as strong momentum raising a score or high volatility lowering it.

Using the same rules for every stock addresses both fragmented information and inconsistent comparison. The calculations are repeatable, and users can trace a result to visible factor values. This approach also provides a clear foundation for testing. Its main weakness is limited interpretation. Fixed templates may explain common numerical results, but they may struggle with mixed news, unusual events, or several signals that point in different directions. The rule-based platform is therefore a strong system foundation and validation baseline, but it provides less support for the interpretation problem.

3.4.3 AI-Assisted Research and Ranking Platform

An AI-assisted research and ranking platform combines integrated data and a documented scoring method with a grounded AI explanation. The system first retrieves the market and news data and calculates and normalizes the ranking. AI then explains the completed score breakdown and its main drivers in plain language. The comparison dashboard allows users to inspect the result, the contributing factors, and the linked citations in one place.

These functions are intended to address all three user problems. The platform brings scattered information together, applies the same comparison method to every stock, and explains the result in language that responds to the specific evidence. For example, an explanation could describe how positive momentum and higher volatility contribute in opposite directions under the chosen scoring rules. It could discuss a relevant news item as context without implying that the item changed the score unless a documented news factor actually did so. The ability to connect several types of evidence is its main advantage over fixed templates.

AI support also requires controls. Even a clear explanation could misstate a news item or conflict with the calculated factor values. StockLens must provide citations, keep calculation and normalization separate from generated text, and ground the explanation in the score breakdown, top drivers, and retrieved evidence. These controls add development work, but they allow the selected solution to provide flexible explanations while keeping the comparison traceable. Established explanation methods such as LIME and SHAP provide useful precedents for attributing model outputs, while StockLens must still test whether its own explanations are faithful and useful [8, 9]. The planned grounding, auditing, and fallback controls also follow the risk-management emphasis of the NIST AI RMF [10].

3.4.4 Automated Investment or Trading System

An automated investment or trading system extends research into portfolio decisions or trade execution. It could recommend how much to buy or sell, connect to a brokerage service, or place orders after analyzing the available information. Such functions may reduce the user's work after research, but they

require account connections, transaction controls, security protections, and much greater responsibility for financial outcomes.

Adding trade execution would not solve the identified research problems more directly than an analysis platform. It would also move beyond the StockLens purpose of helping beginners compare and understand stocks. The stated project scope treats StockLens as a decision-support and educational tool, not an automated trading system or a source of investment advice. Automated trading is therefore excluded from the project scope.

3.4.5 Comparison

Table 9 compares the four solution classes. The entries explain their expected fit with the identified problems and the stated project scope.

Table 9: Comparison of Alternative Solution Classes

Solution class	Main contribution	Main limitation	Decision
Information aggregation dashboard	Brings market and news information into one interface.	Leaves the comparison method and interpretation to the user.	Not selected; meets only part of the need.
Rule-based screening and ranking	Provides repeatable scores with visible rules and factor values.	Has limited ability to explain complex qualitative evidence.	Used as the system foundation and validation baseline.
AI-assisted research and ranking	Combines integrated evidence, consistent ranking, and flexible explanation.	Requires evidence checks and controls for generated content.	Selected StockLens solution class.
Automated investment or trading	Acts on research through recommendations or trade execution.	Exceeds the educational scope and adds major responsibility.	Excluded from project scope.

Table 10 makes the need-level trade-off explicit. Strong means the proposed class directly supports the need; Moderate means support depends on added controls or user effort; Weak means the class does little to address it within this project’s scope. These are design judgments, not measured performance ratings. Complexity is relative to the other concepts, and no numerical total is implied.

Table 10: Qualitative fit of alternatives to needs and course delivery.

Concept	N-01 Comparison	N-02 Understanding	N-03 Evidence	N-04 Feasibility	Complexity
Aggregation dashboard	Moderate	Weak	Strong	Strong	Low
Rule-based baseline	Strong	Moderate	Strong	Strong	Moderate
AI-assisted concept	Strong	Strong*	Strong*	Moderate	Higher
Automated trading	Moderate	Weak	Moderate	Weak	Highest

The dashboard can expose sources but supplies no common scoring rule. The rule-based baseline is easier to reproduce and audit. The selected AI-assisted concept retains that baseline and adds explanations tailored to mixed evidence. Its starred ratings depend on faithful use of supplied scores, dates and citations; they must be tested against the baseline, not assumed from fluent text. Its feasibility is Moderate because provider access, explanation auditing and integration add work. Automated trading receives Weak feasibility because execution controls and account integration exceed the research-only scope; its additional functions do not directly improve the defined comprehension task.

The trade-off becomes clearer when the alternatives are viewed through the user’s journey. An aggregation dashboard helps the user find information, but the user still has to invent a comparison

method and interpret conflicting signals. A rule-based platform solves the consistency problem because the calculation can be repeated and audited, yet fixed explanations may not make complex news understandable. The AI-assisted class keeps that dependable rule-based foundation and adds a flexible explanation layer. Automated trading goes further than the identified need and introduces execution responsibilities that do not help the defined learning task.

StockLens therefore selects the AI-assisted research and ranking platform, but the selection is conditional rather than automatic. Data retrieval and the numerical ranking occur first; the AI receives the completed score breakdown, top drivers, dates, and retrieved evidence only to explain the result. The dashboard lets the user inspect both the calculation and its sources. This arrangement preserves a checkable baseline while allowing the team to test whether the explanation actually improves understanding. The N-02 task and explanation audit must be compared with the rule-based baseline. If the AI introduces contradictions, misses the proposed comprehension criterion, or requires resources the team cannot confirm, StockLens should retain the rule-based comparison and revise the explanation approach instead of claiming an unmeasured benefit.

4 Assumptions

4.1 Assumptions and Confirmation

The StockLens concept depends on several conditions that have not yet been demonstrated. Recording them as assumptions keeps the team from treating a convenient idea as established fact. These assumptions are different from choices the team controls, such as the 100-stock boundary, and from targets that still require testing, such as an 80% task-success rate. Each assumption needs a named owner, a confirmation method, an evidence location, and a review date; until that evidence is reviewed, the assumption remains open.

The confirmation order follows the operating story. First, the team must show that suitable sources are accessible and that their inputs can be aligned (A-01 and A-04). Next, it must confirm that the external AI service can accept the intended evidence and return usable outputs (A-02). Only then does it make sense to demonstrate the nominal workflow (A-05) and evaluate it with intended users (A-03). If an assumption fails, the affected needs, interfaces, and model views should be revised before the concept is treated as the shared baseline.

Table 11 turns each uncertain premise into a concrete check and states what the team should do if the evidence does not support it.

Table 11: Assumptions, affected needs and actions for confirmation.

Assumption	Reason / confirmation activity	Response if unsupported
A-01 Suitable market and news sources are accessible. <i>N-01, N-03, N-04, N-07</i>	The market/news retrieval function needs usable daily records, company identifiers and dated news evidence. Check selected sources for coverage, sample records, source identifiers and permitted use.	Narrow the eligible universe or change providers; reassess what may be displayed or retained. Do not promise full coverage before access is confirmed.
A-02 AI analysis can use an external service. <i>N-02, N-04, N-08</i>	Model hosting is outside the SOI; the StockLens AI Explanation Module coordinates requests. Check service access, resource limits, allowed inputs and representative returned outputs.	Reconsider the provider or AI approach. If a local model is required, revisit the SOI boundary, hosting resources and relevant model entities.
A-03 Intended users benefit from daily comparisons. <i>N-01–N-03</i>	Daily observations bound the workload and suit the proposed learning task. Use the comparison pilot and follow-up feedback to check whether this frequency supports users' actual research needs.	Revisit the target user segment or refresh requirement. Do not silently extend the concept to intraday trading.
A-04 Scores can be computed from aligned inputs. <i>N-01, N-02</i>	The scoring path requires compatible symbols, observation periods, news relevance and a defined scoring configuration. Check alignment, normalization, weights and the provenance of each scoring input on representative cases.	Revise preprocessing, scoring rules or stock eligibility before interpreting ranks. Recheck whether the explanations still match the revised calculation.
A-05 UC.1 starts with valid requests and normally responding sources and services. <i>N-04; bounds N-01–N-03 validation</i>	The initial behavioral model covers one nominal path. Its Walking Skeleton uses fixed stubs to check connectivity; real provider access and input quality require separate evidence.	Add separate exception scenarios in later work.

4.2 Scope Constraints and Unvalidated Targets

Assumption: an uncertain premise, such as usable provider access (A-01), that needs confirmation.

Chosen scope constraint: up to 100 US stocks, daily market observations and a research-only service. Automated trading, order execution, personalized portfolio management and intraday trading support are excluded.

Unvalidated target: a desired result, such as the proposed 80% task-success threshold.

The proposed 80% thresholds in the user MOEs are decision criteria for a future pilot. Record both counts and percentages for each task, use the same dated inputs and rubric, and report assistance separately. Audit source support and agreement with score inputs alongside the user tasks, and report the sample limits.

Confirm scoring and AI settings through the planned evidence checks. Update affected needs, interfaces and model views when assumptions change.

5 Summary

StockLens starts from a practical problem: a beginning investor can find plenty of stock information and still struggle to make a fair comparison or explain a ranking. The proposed solution connects that journey from end to end. It brings dated market and news evidence together, applies one documented comparison method, and presents the result with score drivers, sources, and a plain-language explanation. The user remains responsible for reviewing the evidence and making any investment decision.

This report identifies six stakeholder groups and eight preliminary needs, then connects those needs to proposed measures, validation methods, operational responsibilities, and life-cycle activities. The selected AI-assisted concept is built on a rule-based calculation baseline. AI explains a completed result; it does not independently determine the numerical order. That separation is central to making StockLens both understandable and auditable.

The models and tables describe the intended system, not proof that it already works. Provider access, input alignment, scoring definitions, explanation quality, and the proposed success thresholds still require evidence. Table 7 links the main goals to observable checks, while Table 8 records the shared operational baseline across the model views.

The next step is to turn the assumptions into evidence: confirm provider access and scoring rules, map the fixed-stub implementation to the named model participants, and run the planned user tasks and source audits. Those results will show whether StockLens truly helps users compare, understand, and question the evidence rather than simply giving them another ranking.

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